

Shivang Dubey

LinkedIn: [linkedin.com/in/shivangdubey8](https://www.linkedin.com/in/shivangdubey8)

GitHub: github.com/shivangdubey

Email : shivangdubey2001@gmail.com

Mobile : +91-9717428962

Portfolio: shivangdubey.github.io

SUMMARY

Data Engineer at ZS Associates with **2.5+** years of experience delivering ETL pipelines, analytics platforms, and backend systems for a Fortune 500 pharmaceutical client. Engineered **500+** data pipelines processing 5M+ healthcare records at 99% accuracy, automated workflows saving 24+ hours weekly, and led backend development for a **1,000+** user platform. Currently pursuing a PG Diploma in Data Science in Health and Climate Change from IIIT Delhi, deepening expertise in healthcare data and applied ML.

SKILLS SUMMARY

- **Languages:** Python, SQL (MySQL, PostgreSQL), C/C++
- **Tools & Frameworks:** pgAdmin, Numpy, Pandas, OpenCV, Flask, Git, GitHub
- **BI & Reporting Tools:** Power BI, MicroStrategy, Excel, Powerpoint
- **ETL & Orchestration:** Apache Airflow, IICS
- **Cloud & Data Platforms:** AWS (S3, EC2, ECS, Redshift), Snowflake
- **AI Tools:** Claude, ChatGPT, Gemini (prompt engineering for development & analytics workflows)
- **Key Skills:** Data Analysis, Data Visualization, ETL Development, Data Ingestion, Backend Development, Machine Learning, Healthcare Data (Pharma Drug Data, Claims, Physician & Market Sales Analytics)

EXPERIENCE

- **ZS Associates** Gurugram, India
Business Technology Solutions Associate Consultant Sep 2023 - Present
 - Developed and executed scalable ETL processes that converted **500+** raw datasets into actionable sales and compensation reports, implementing rigorous data validation measures to achieve **99%** data accuracy across systems.
 - Conducted routine data manipulation and insight generation using SQL (PostgreSQL, Workbench). Managed data ingestion and extraction via Amazon S3, and processed large healthcare datasets including medical sales and claims data exceeding **5M+** records.
 - Developed performance analytics dashboards in Power BI and MicroStrategy for a field force of **1,000+** healthcare representatives, enabling individual performance tracking and contributing to a 25% increase in goal attainment.
 - Automated backup processes on Amazon S3 using Python and **boto**, reducing manual effort by approximately **8 hours** per week. Streamlined operations and reporting workflows using Apache Airflow, saving an additional 4 hours weekly.
 - Developed Excel automation solutions using VBA, eliminating a **12-hour** weekly manual process for data fetching, formatting, and analysis. Created VBA-driven Excel reports linked to PostgreSQL databases, enhancing team efficiency by approximately **80%**.
 - Led backend engineering for a web portal as part of a critical technology migration project, implementing complex business logic with Python and Flask and integrating with an AngularJS frontend. The platform supports **1000+** users in a US healthcare marketing ecosystem managing data for **4** key pharmaceutical products.
 - Managed and mentored a team of **4 Associates**, overseeing onboarding, conducting regular knowledge transfer sessions, and providing technical guidance on ETL development, SQL, and data workflows — ensuring consistent delivery standards across the team.
 - Collaborated directly with client stakeholders to gather requirements, translate business needs into technical specifications, and deliver reporting solutions aligned to sales and compensation workflows.
- **Polytechnique Montréal** Montréal, Canada
Summer Research Intern May 2022 - Jul 2022
 - Collaborated with a PhD student to develop a **Deep Q Learning** model for a network intrusion detection system. The obtained accuracy was greater than **79%**.
 - Developed a Deep Q-Learning model for network intrusion detection, outperforming SVM, Random Forest, and CNN baselines by **2%** accuracy using TensorFlow.

EDUCATION

- **Indraprastha Institute of Information Technology Delhi** IIIT Delhi
Post Graduate Diploma in Data Science in Health and Climate Change for Social Impact Oct 2025 - Present
- **Guru Gobind Singh Indraprastha University** BPIT Delhi
B.Tech. in Electronics and Communication Engineering; CGPA: 9.45/10.00 Aug 2019 - Jul 2023

ACADEMIC PROJECTS

- **Social Distancing Violation Detection System:** The model was created to detect human and distance violations between two human centroids along with human face detection.
 - **Tech:** Model trained using **YOLO** (v3) algorithm and weights, along with COCO dataset for pre-existing object classes. **OpenCV** for face and human detection. Along with **SciPy**'s Euclidean Distance for human centroid formation.
 - **Outcome:** With a suitable detection rate, the model draws a box around the human; Accuracy more than **94%**. [Thesis](#)